

Fundamentals of Operating System Design

Module 8: User Address Space and Kernel Address Space

Prashant Jadav
DEPSTAR-CHARUSAT

Learning Outcomes

- Explain process address space.
- Differentiate user and kernel address spaces.
- Understand process memory layout.
- Explain virtual and physical addresses.

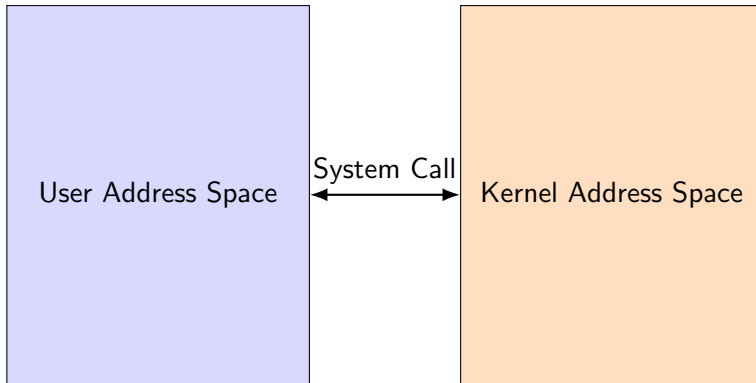
What is an Address Space?

Definition

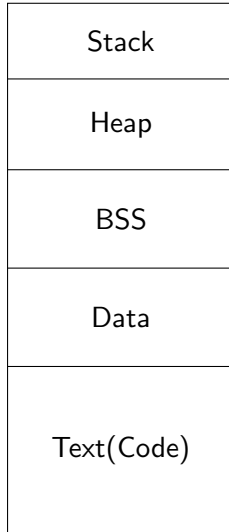
An address space is the range of memory addresses that a process can use during execution.

- Every process has its own virtual address space.
- The operating system protects one process from accessing another process's memory.

User Address Space vs Kernel Address Space



Process Memory Layout



Segments of a Process

Segment	Purpose
Text	Executable instructions
Data	Initialized global variables
BSS	Uninitialized global variables
Heap	Dynamic memory allocation
Stack	Function calls and local variables

Virtual Address vs Physical Address

Virtual Address	Physical Address
Generated by CPU	Actual RAM location
Visible to program	Managed by MMU
Each process has its own	Shared physical memory
Mapped by OS	Stored in RAM

Why Separate Address Spaces?

- Memory protection
- Prevent application crashes from affecting the kernel
- Better security
- Process isolation
- Controlled access through system calls

Example

Scenario

A text editor crashes because of a programming bug.

Because it runs in the user address space,

- Other applications continue running.
- The kernel remains protected.
- The operating system does not crash.

Summary

- Every process has its own virtual address space.
- User address space is isolated from kernel address space.
- System calls provide controlled access to kernel memory.
- Virtual addresses are translated to physical addresses.
- Process memory consists of Text, Data, BSS, Heap and Stack segments.

Quick Review

- ① What is an address space?
- ② Differentiate user and kernel address spaces.
- ③ Draw the process memory layout.
- ④ Differentiate virtual and physical addresses.
- ⑤ Why is memory protection required?